

1. Administrative & Study Identification

Full Study Title: Real-life Treatment Outcomes of Ravulizumab in Polish Patients With Paroxysmal Nocturnal Hemoglobinuria (PNH). Prospective and Retrospective, Multicenter, Non-interventional Study **Short Title/Acronym:** PNH-RECORD / Real-life Treatment Outcomes of Ravulizumab in PNH **Protocol Number:** D9289R00001 **NCT Number:** NCT06633536 **Posting ID:** 257780608235168464 **Trial Status:** Active, Not Recruiting (ACTIVE_NOT_RECRUITING) **Sponsor Name:** AstraZeneca **Investigational Product:** Ravulizumab (Ultomiris) **Phase of Development:** Not Applicable / Observational (Non-interventional) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To collect data on the patients' characteristics and clinical outcomes of ravulizumab administered in the scope of routine clinical practice in PNH. **Secondary Objectives:** To evaluate real-world effectiveness parameters, including lactate dehydrogenase (LDH) levels, incidence of breakthrough hemolysis, hemoglobin stabilization, transfusion avoidance, and real-life safety profiles.

2.2 Background & Rationale To evaluate the real-life treatment outcomes of ravulizumab in Polish patients with Paroxysmal Nocturnal Hemoglobinuria (PNH) receiving ravulizumab treatment within the framework of the National Drug Program (NDP) in Poland. While ravulizumab has proven efficacy in controlled clinical trials, there is a medical necessity to collect real-world data retrospectively and prospectively to understand its long-term characteristics, effectiveness, and safety in routine clinical practice.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Non-interventional), retrospective with prospective follow-up **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Observational cohort)

3.2 Planned Enrollment & Duration Target \$N\$: 64 patients **Number of Sites:** Multicenter (Poland) **Estimated Study Start:** 06-Feb-2025 **Estimated Primary Completion:** 01-Aug-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult (aged ≥ 18 years) patients with PNH receiving ravulizumab treatment in the frames of NDP in Poland. 2. Patients willing to participate in the study and signed Informed Consent Form (ICF).

4.2 Exclusion Criteria 1. Those who participated in a ravulizumab clinical trial in the past, and/or those who participated/plans to participate in a clinical trial on/after the date of first ravulizumab infusion through NDP. 2. Cognitive incapacity, unwillingness, or language barriers precluding adequate understanding or cooperation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ravulizumab administered according to routine clinical practice guidelines and the Polish National Drug Program (NDP) protocol. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Long-term routine treatment evaluated retrospectively and prospectively up to the study completion date (Estimated August 2028).

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications and supportive care for PNH allowed under routine clinical practice. **Prohibited:** Concurrent participation in other interventional clinical trials from the date of the first ravulizumab infusion through the NDP.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening/Baseline	N/A Informed Consent, extraction of baseline characteristics and historical retrospective clinical data via eCRF
2	Prospective Follow-up	Routine Care Data collection during standard clinical visits/infusion appointments (Efficacy, Safety, LDH labs)
3	End of Study	Aug 2028 Final prospective data extraction, overall outcomes assessment

(Note: Because this is an observational non-interventional study, the visit schedule aligns strictly with the patients' routine clinical care pathways rather than protocol-mandated days.)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Change in LDH levels from index date (first dose of ravulizumab) at every 6 month. 2. Proportion of patients achieving or remaining at LDH<1.5 ULN at

every 6 month. **Measurement Method:** Retrospective and prospective data extraction from patient medical records / eCRF tracking.

7.2 Secondary & Exploratory Endpoints 1. Proportion of transfusion-free patients from index date at every 6 month. 2. Number of blood units used per patient-year. 3. To present patients characteristics.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collected during routine clinical practice and abstracted into the study eCRF. **Serious Adverse Events (SAEs):** Reported per standard post-marketing pharmacovigilance and observational study guidelines. **Data Monitoring Committee (DMC):** Sponsor internal pharmacovigilance and observational research oversight.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All enrolled eligible patients receiving \$ \ge 1\$ dose of ravulizumab via the NDP who provide informed consent. **Sample Size Justification:** Target \$N = 64\$ represents the estimated real-world cohort size eligible under the Polish National Drug Program. **Handling of Missing Data:** Standard handling for observational studies; generally analyzed as observed without data imputation to accurately reflect real-world gaps.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP), Good Clinical Practice (GCP), and the Declaration of Helsinki. **Monitoring Plan:** Periodic data verification of eCRF entries against electronic health records/source documents at participating Polish sites. **Audit Trail:** Secure, validated eCRF system utilizing strict audit trails and data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: D9289R00001 **Registry Number:** NCT06633536 **Posting ID:** 257780608235168464 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** PNH-RECORD **Official Regulatory Title:** Real-life Treatment Outcomes of Ravulizumab in Polish Patients With Paroxysmal Nocturnal Hemoglobinuria (PNH). Prospective and Retrospective, Multicenter, Non-interventional Study

12. Clinical Framework (PNH Specific)

Primary Condition: Paroxysmal Nocturnal Hemoglobinuria (PNH)
Diagnosis Threshold: Confirmed PNH diagnosis requiring complement C5 inhibitor therapy under the NDP **Medical Methodology:** Complement C5 Inhibition (Ravulizumab) **Optimization Target:** Control of intravascular hemolysis (LDH normalization) and prevention of breakthrough hemolysis

13. Study Procedures & Methodology

Management Pathway: Standard routine clinical pathway for PNH within the Polish National Drug Program **Education Protocol (HCP):** Standard NDP prescribing guidelines and real-world data extraction training **Education Protocol (Patient):** Routine clinical practice counseling for PNH treatment **Quality Audit Mechanism:** eCRF source data verification and observational tracking

14. Observation & Follow-Up Schedule

Total Duration: Retrospective look-back + Prospective follow-up up to \$\\sim\$ 42 months (Feb 2025 to Aug 2028) **Onsite Visit Cadence:** Aligned with routine clinical infusion and monitoring schedule **Remote Monitoring Frequency:** Data-driven eCRF monitoring continuously **Checklist Review Interval:** Periodic data extraction intervals as defined by the observational protocol

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				